

Master project: Bridging Vision and Language: A Multi-Image AI Pipeline for Context-Aware NPC Storytelling

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1 Introduction

The landscape of interactive media and video games has evolved from simple, linear progression into expansive, open-world environments where player agency dictates the flow of events. As rendering technologies and physics engines have advanced to create highly realistic simulations, the systems governing in-game storytelling have often lagged behind. In most contemporary open-world games, Non-Playable Character (NPC) dialogue remains heavily pre-scripted. While branching dialogue trees offer the illusion of choice, they are ultimately finite and struggle to account for the emergent, unpredictable actions players take in a physics-driven world.

2 Abstract

Modern interactive media frequently struggles with ludonarrative dissonance, where dynamically driven player actions conflict with static, pre-scripted game narratives. While recent advancements in Vision-Language Models (VLMs) have enabled narrative extraction from single, static images, these approaches lack the temporal and sequential context required to interpret complex gameplay. This project introduces a novel, multi-modal machine learning pipeline designed to generate context-aware, dynamic dialogue for Non-Playable Characters (NPCs) based on sequential gameplay screenshots. The proposed architecture separates visual perception from narrative generation to strictly ground the AI in gameplay reality. First, a VLM sequentially processes in-game images to extract key entities, actions, and environmental states, structuring this data into a chronological scene graph. Second, a Large Language Model (LLM) utilizes this structured data—constrained by established interactive storytelling principles—to generate dramatically meaningful and credible NPC dialogue. By filtering raw visual data into a strict JSON framework before text generation, the pipeline prevents LLM hallucination and arbitrary content creation. This

approach demonstrates a scalable method for bridging computer vision and natural language processing, allowing games to dynamically translate visual player events into coherent, immersive lore without sacrificing narrative integrity.

3 Code will be pasted here